

CHINTAN VAGHAMSHI

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EDUCATION

Indian Institute of Technology Jodhpur

B.Tech in Electrical Engineering | CGPA: 7.50/10.00

Ashadeep Vidhyalaya

Intermediate | Percentage : 87%

Ashadeep Vidhyalaya

Matriculation | Percentage : 95%

Aug 2024 – Jul 2028

Jodhpur, Rajasthan

May 2022 – May 2024

Surat, Gujarat

May 2020 – May 2022

Surat, Gujarat

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS

Technologies: Node.js, React, Express, MongoDB, Redis, BullMQ, Socket.io, Microservices, REST APIs, WebSockets, JWT, WebCrypto API

Tools: Git, Vercel, Render, Cloudflare R2, Cloudinary, Postman, VS Code

EXPERIENCE

Open Source Contributor | NetworkX

Nov 2025 – Dec 2026

Python, Graph Theory, Open Source

Remote

- Analyzed the NetworkX graph-processing codebase, focusing on algorithm correctness across graph traversal APIs.
- Enhanced documentation for core graph APIs by developing a new reference page for reporting views, integrated into the main documentation site; reduced onboarding time for new contributors by 30%.

PROJECTS

Lithium-matcher | [Github](#) | [Video Explanation](#) | C++20, Python (Synthetic Data Generation), GCC / MinGW, CMake

- Engineered a C++20 matching engine achieving a peak threaded throughput of **17.2M orders/sec**, with a core **p50 latency of 28ns** and **p99 of 150ns** (verified via `–O3` optimized hardware cycle counter benchmarking).
- Developed a 240MB fixed-capacity Memory Pool and an **O(1) Limit Order Book**, yielding a **32x allocation speedup** and guaranteeing zero-copy, in-place updates.
- Implemented a lock-free SPSC ring buffer to decouple ingestion from matching; resolved a critical wraparound aliasing bug in buffer-full detection that caused silent order drops.
- Devised the Limit Order Book using `std::array` indexed directly by price levels to achieve strict **O(1) time complexity** for all order insertions and matches, completely eliminating the **O(log N)** traversal overhead typical of `std::map` implementations.
- Profiled the full pipeline in isolated stages across 3M Python-generated synthetic orders, separating matching engine cost from I/O overhead to avoid conflated measurements and isolate true matching engine performance.

SecureShare | [Github](#) | [Live](#) | [Video Explanation](#) | JavaScript, React, Node.js, Express, MongoDB, Cloudinary, WebCrypto API

- Designed a **zero-knowledge E2E encrypted** file sharing platform across **18 REST APIs** — browser encrypts files via **WebCrypto (AES-256-GCM + RSA-2048 OAEP)** before upload; **server mathematically incapable of reading user files** even under full infrastructure compromise.
- Implemented **PBKDF2 key derivation** (100K iterations), MongoDB-persisted account lockout, **JWT refresh token rotation** with HttpOnly cookies, and **SHA-256 integrity verification** — defeating **4 simultaneous attack vectors** (XSS, CSRF, brute force, tampering).
- Established a complete security audit trail across upload, download, share, revoke, and auth events, with a user dashboard surfacing real-time failed login attempts and **INTEGRITY_FAIL** alerts — **zero plaintext file exposure** verified across MongoDB, Cloudinary, and server RAM.

WayPoint | [Github](#) | [Live](#) | JavaScript, React, Node.js, Express, MongoDB, Redis, BullMQ, Socket.io, Cloudflare R2

- Spearheaded a real-time collaborative platform (MERN) with Socket.io.
- Refactored task-update endpoint to use a non-blocking 'Fire-and-Forget' background write pattern, reducing server Time To First Byte (TTFB) from **2.33s to 1.10s (53% reduction)**.
- Engineered JWT auth with access/refresh token rotation and 4-tier RBAC (owner/admin/member/viewer) across **36 REST endpoints**; integrated collaborative document editing using **Y.js (CRDTs)** for conflict-free multi-user sync.
- Designed Redis (Upstash) caching layer with event-driven invalidation (**85% hit rate**); tuned BullMQ worker polling (1s→30s), cutting idle DB requests **30x (170k→5.7k/day)** to drastically reduce infrastructure costs.

COURSEWORK

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|--------------------------------|----------------------------|--|
| • Data Structures & Algorithms | • Graph Theory | • Pattern Recognition & Machine Learning |
| • Digital Design | • Probability & Statistics | • Artificial Intelligence |

ACHIEVEMENTS & PARTICIPATION

- Solved **300+** DSA problems across LeetCode and competitive programming platforms.
- Achieved **Gold Medal** In SOF Science Olympiad
- Participated in **Smart India Hackathon** (national-level government hackathon).
- Ranked **3rd in State** in Karate; competed at **National Level twice**.

LEADERSHIP

- **Core Member**, E-Cell, IIT Jodhpur — organized and managed entrepreneurship events over 9 months.
- **Core Member**, Programming Club, IIT Jodhpur — co-organized competitive programming contests and workshops.
- **Core Member**, SWC IIT Jodhpur, , Mentored 10 freshers academically and non-academically.